Unit 6 · Session 1 · Tue, Nov 24, 2026

Genitourinary & Reproductive Pharmacology — Session 1

3 parts · every fact cited to its slide or textbook page. About 100 minutes of focused study.

Study plan

15 min	**Read the guide**	One pass, top to bottom. Bold = what the lecture emphasized.
50 min	**Diagrams & tables**	Redraw each flow from memory; cover a table column and recall it.
10 min	**Cram sheet**	Hide answers and quiz yourself. Repeat until it's automatic.
20 min	**Self-test**	Answer before you look. Read every explanation, even when right.
5 min	**Objectives check**	Tick each objective only if you can teach it out loud.

HOW TO READ THIS GUIDE

- **Bold** = emphasized on the lecture slides
- **TB** textbook only · **SG** course study guide only
- S12 = slide 12 · S12 (notes) = speaker notes · SG U1-R3 = study guide item · p. 17 = Katzung page
- Open **Sidekick** (bottom right) to search, listen, drill flashcards, quiz, or chat about this lecture

The whole lecture in 6 sentences

1. **Estrogens** bind ER-α/β (VTE risk with smoking, age > 35); **unopposed estrogen** with a uterus → endometrial cancer, so add a **progestin.** S5; S6; SG U6-R1; SG U6-R2

2. Estrogen contraindicated (VTE, smoker > 35, migraine with aura) → **progestin-only** is the right first choice; **DMPA** = reversible bone loss. S9; SG U6-R3; SG U6-R4

3. **SERMs**: tamoxifen blocks breast but stimulates endometrium (new bleeding → evaluate); raloxifene is endometrium-neutral; both raise VTE. S11; S18; SG U6-R5

4. **Aromatase inhibitors** block peripheral estrogen synthesis — **postmenopausal only**; bone loss and joint pain. S13; S14; SG U6-R6

5. **Testosterone**: erythrocytosis (hematocrit), PSA; **antiandrogens** block the receptor; **5-ARIs** block DHT formation. S15; S16; SG U6-R7; SG U6-R8

6. **PCOS**: hyperinsulinemia ↑ ovarian androgens and ↓ SHBG; **metformin** works indirectly and is usually combined with contraception. S20; S22; S25; SG U6-R9; SG U6-R10

Learning objectives

IO = syllabus unit objective · SO = this session's objectives slide.

IO6.1 Identify major pharmacologic classes used in the management of genitourinary and reproductive conditions and describe their mechanisms of action.

Part 1 · Estrogens, Progestins & Contraception
Part 2 · SERMs, Aromatase Inhibitors & Androgens
Part 3 · PCOS & the Reproductive-Metabolic Overlap

IO6.2 Describe the indications, contraindications, adverse effects, and clinically significant interactions associated with medications used to treat genitourinary and reproductive conditions.

Part 1 · Estrogens, Progestins & Contraception
Part 2 · SERMs, Aromatase Inhibitors & Androgens

IO6.3 Compare pharmacologic agents within and across therapeutic classes to distinguish clinically relevant differences in efficacy, safety, and use.

Part 2 · SERMs, Aromatase Inhibitors & Androgens
Part 3 · PCOS & the Reproductive-Metabolic Overlap

IO6.4 Select appropriate pharmacologic therapy for common genitourinary and reproductive conditions based on clinical presentation and patient-specific factors.

Part 1 · Estrogens, Progestins & Contraception
Part 3 · PCOS & the Reproductive-Metabolic Overlap

IO6.5 Identify appropriate dosing, administration, and monitoring considerations for medications used in genitourinary and reproductive conditions.

Part 2 · SERMs, Aromatase Inhibitors & Androgens
Part 3 · PCOS & the Reproductive-Metabolic Overlap

IO6.6 Explain important medication-use and safety information to patients receiving genitourinary and reproductive pharmacotherapy.

Part 1 · Estrogens, Progestins & Contraception

SO1 Describe the physiologic roles and clinical uses of estrogens and progestins, including combined and progestin-only hormonal contraception.

Part 1 · Estrogens, Progestins & Contraception

SO2 Describe the mechanism, tissue-selective actions, and adverse effects of selective estrogen receptor modulators (SERMs) and aromatase inhibitors.

Part 2 · SERMs, Aromatase Inhibitors & Androgens

SO3 Describe the clinical uses, adverse effects, and monitoring considerations for androgens and antiandrogens.

Part 2 · SERMs, Aromatase Inhibitors & Androgens

SO4 Explain the pathophysiology of polycystic ovary syndrome (PCOS) as a reproductive-metabolic overlap condition, including the central role of insulin resistance.

Part 3 · PCOS & the Reproductive-Metabolic Overlap

SO5 Describe the role of metformin and other antidiabetic agents in the management of PCOS and related reproductive-metabolic conditions.

Part 3 · PCOS & the Reproductive-Metabolic Overlap

ABOUT THIS LECTURE

- **PA PHARMACOLOGY COURSE · LECTURE 6 · SESSION 1 OF 2** — Genitourinary & Reproductive Pharmacology. **Session 1: Gonadal Hormones and the Reproductive-Metabolic Overlap | Tuesday, November 24 | 8:30–11:00 AM** S1
- Organizing idea: the hypothalamic-pituitary-gonadal axis. Every drug in this session replaces a sex steroid, blocks its receptor, blocks its synthesis, or reduces the metabolic driver, insulin resistance, that fuels androgen excess in PCOS. S1 (notes)
- Arc: gonadal steroid pharmacology first (estrogens, progestins, SERMs, aromatase inhibitors, androgens/antiandrogens), then the reproductive-metabolic overlap through PCOS. S2 (notes)
- Preview: autonomic pharmacology of the bladder and lower urinary tract continues in Session 2 (Tuesday, December 1) — mechanistically distinct, but sharing cross-system classes (alpha1 blockers, NO-cGMP signaling). S2; S2 (notes)
- Readings: Katzung chapter 40 (The Gonadal Hormones and Inhibitors) and selected sections of chapter 41 (Pancreatic Hormones & Antidiabetic Drugs) focused on the reproductive/metabolic overlap (e.g., PCOS and insulin resistance). S3; S3 (notes)
- Chapter 6 (Introduction to Autonomic Pharmacology) — the sections on Autonomic Receptors and Non-Adrenergic, Noncholinergic (NANC) Neurons — supports Session 2's bladder and lower urinary tract content, not today. S3; S3 (notes)
- Required text: Katzung's Basic and Clinical Pharmacology, 16th Edition (Todd W. Vanderah, 2024/2025, McGraw-Hill Education; ISBN 1260463303 eBook). Pharmacotherapy Principles and Practice (Chisholm-Burns et al., 2022) remains additional support alongside all assigned chapters. S3
- **Appendix 1** — Drug Profiles: study reference for the drugs covered in Session 1. Know the dose logic, the serious adverse effect, the contraindication that changes decisions and the toxicity you would need to recognize. S28; S28 (notes)
- **Abbreviation key PO = by mouth IM = intramuscular IV = intravenous SC = subcutaneous BID / TID = twice / three times daily ER = extended-release IUD = intrauterine device DMPA = depot medroxyprogesterone acetate** S28; S28 (notes)
- **GnRH = gonadotropin-releasing hormone VTE = venous thromboembolism DVT / PE = deep vein thrombosis / pulmonary embolism MI = myocardial infarction PCOS = polycystic ovary syndrome BPH = benign prostatic hyperplasia** S28; S28 (notes)
- **PSA = prostate-specific antigen CBC = complete blood count CYP = cytochrome P450 GI = gastrointestinal DCIS = ductal carcinoma in situ eGFR / CrCl = estimated glomerular filtration rate / creatinine clearance ER-positive = estrogen receptor-positive** S28; S28 (notes)
- **Standard adult dosing for teaching; verify against current prescribing information.** S29-44
- **Appendix 2: DRUG PROFILE · SUPPORTING** — supporting drugs explain interactions. Listed for context; not profiled in detail. S45; S45 (notes)
- **Original figures in this deck:** Figures 2, 4 and 7 (The Hypothalamic-Pituitary-Gonadal (HPG) Axis: Sites of Hormonal Drug Action; Two Ways to Reduce Estrogen Signaling: SERMs vs. Aromatase Inhibitors; The PCOS–Insulin Resistance Cycle) were generated by the course director using the AI tool Claude. S46
- They are original diagrams, not reproduced from Katzung's Basic and Clinical Pharmacology or any other copyrighted source. **Mechanism illustrations:** Figures 1, 3, 5, 6 and 8 are illustrative AI-generated schematics. **Reference:** Anthropic. (2026). Claude (Sonnet 5) [Large language model]. https://claude.ai S46; S46 (notes)

Slide discrepancies: none found (slides, speaker notes, and study guide agree).

1 PART 1 · DRUG CLASS

Estrogens, Progestins & Contraception

SO1 IO6.1 IO6.2 IO6.4 IO6.6

CLASS

- **The two core gonadal steroid classes, and how they are combined for contraception and hormone therapy.** S4
- **Figure 2. Original illustrative diagram (not from Katzung): GnRH drives FSH/LH release, which drives gonadal sex-steroid production, which in turn negatively feeds back on the hypothalamus and pituitary —** S8
- **with four distinct classes of drugs intervening at different points along this axis.** This is the organizing map for the session: GnRH agonists/antagonists, SERMs, aromatase inhibitors and antiandrogens each anchor to a point on it. S8; S8 (notes)

PROTOTYPE + OTHER DRUGS NAMED

- Estradiol and ethinyl estradiol (estrogens); norethindrone, medroxyprogesterone, levonorgestrel (progestins); combined oral contraceptives. S5; S6; S26

MECHANISM OF ACTION

- **Mechanism: bind intracellular estrogen receptors (ER-alpha, ER-beta), regulating gene transcription in reproductive, bone, cardiovascular, and CNS tissue.** S5; S4 (notes); SG U6-R1
- Estrogen physiologic effects: endometrial proliferation, breast development, bone density maintenance, favorable lipid profile effects (↑ HDL, ↓ LDL). S5; SG U6-R1
- **Mechanism: bind progesterone receptors, counteracting estrogen-driven endometrial proliferation and thickening cervical mucus.** S6

INDICATIONS

- Estrogen clinical uses: hormonal contraception (combined with a progestin), menopausal hormone therapy for vasomotor symptoms, and hypogonadism. S5
- Progestins: a required component of combined hormonal contraception (protects the endometrium from unopposed estrogen), progestin-only contraception, and management of abnormal uterine bleeding. S6

- Unopposed estrogen chain (Figure 1): estrogen binds ER in the endometrium → proliferation and thickening of the lining → without progestin: hyperplasia and cancer risk → progestin binds PR: limits proliferation, matures the endometrium. S7; S7 (notes)
- Unopposed estrogen drives unchecked endometrial proliferation; a progestin counteracts this proliferation. SG U6-R2
- Combined methods (oral pills, transdermal patch, vaginal ring) suppress ovulation via negative feedback on FSH/LH. S9
- **TB** Estrogens raise hepatic clotting factors II, VII, IX and X and lower antithrombin III — the basis of the clotting risk; transdermal or vaginal routes avoid first-pass liver exposure. p. 761; p. 762

- Progestin-only methods — the "mini-pill," depot medroxyprogesterone acetate (DMPA) injection, the subdermal implant, and the levonorgestrel intrauterine system — avoid estrogen-related VTE risk entirely. S6; S9; SG U6-R4
- Combined methods are often chosen for added benefits: cycle regulation, acne/hirsutism improvement. S9
- When estrogen is contraindicated, a progestin-only method is not a compromise — it is often the correct first choice (safe with VTE history, migraine with aura, or smokers over 35). S9; S9 (notes)
- **TB** Postmenopausal hormonal therapy: vasomotor symptoms, genital atrophy and bone loss; after hysterectomy estrogen alone is fine because no progestin is needed. p. 762; p. 763
- **TB** An implantable progestin preparation (subdermal implant) provides long-acting progestin-only contraception. p. 768
- **TB** Contraception with progestins alone suits patients with hepatic disease, hypertension, or prior thromboembolism. p. 773

ADVERSE EFFECTS (BOXED WARNINGS FIRST)

- **Adverse effects: venous thromboembolism (VTE) risk, increased risk with smoking and age over 35, nausea, breast tenderness, and a possible increase in certain hormone-sensitive cancer risks with long-term use.** S5
- Estrogen's VTE risk is the single most tested adverse effect: know the risk factors (smoking, age > 35, personal/family VTE history) that shift a patient from "candidate" to "contraindicated." S5 (notes)

CONTRAINDICATIONS

- Estrogen contraindications: history of VTE, estrogen-sensitive breast cancer, uncontrolled hypertension, and smokers over age 35 — all raise cardiovascular/thrombotic risk substantially. Smoking combined with age over 35 makes a progestin-only method (no estrogen, no added VTE risk) the appropriate choice. S5; S5 (notes); SG U6-R3
- Combined methods carry estrogen-related VTE risk — require screening for contraindications. Hysterectomy patients can take estrogen alone. S7; S9

- **Adverse effects: irregular bleeding/spotting (especially early in use), mood changes, and — with DMPA specifically — reversible bone mineral density loss with prolonged use.** S6; S9; SG U6-R4
- **Unopposed estrogen (estrogen without a progestin) in a patient with a uterus significantly increases endometrial cancer risk — this is why progestin-only or combined regimens, never estrogen alone, are used in patients with an intact uterus.** S6; SG U6-R2
- **TB** Oral contraceptive VTE risk tracks the estrogen, not the progestin; factor V Leiden and other inherited clotting disorders markedly raise it. p. 772
- **TB** Estrogen therapy also increases migraine, cholestasis, gallbladder disease and hypertension; oral contraceptives raise hepatic adenoma risk. p. 764; p. 772

- **TB** Contraindications & cautions: thrombophlebitis or past thromboembolism, cardiovascular or cerebrovascular disease, estrogen-dependent cancer, undiagnosed genital bleeding, liver disease. p. 764; p. 773

DRUG INTERACTIONS

- Enzyme inducers (rifampin, carbamazepine, phenytoin, St. John's wort) lower combined contraceptive efficacy. S45; S45 (notes)
- **TB** Rifampin and phenytoin speed estrogen/progestin metabolism, and antibiotics that disturb gut flora may cut enterohepatic recycling — both can cause contraceptive failure. p. 771; p. 773

PHARMACOKINETICS

Not covered in materials

MONITORING

- Progestin-only methods have more irregular bleeding patterns, especially initially; DMPA specifically: reversible bone density loss with long-term use. S9

CLINICAL PEARL / WHY

- The "unopposed estrogen" point is a classic exam item: any patient with a uterus receiving estrogen needs a progestin too, unless she has had a hysterectomy. S6 (notes); S7 (notes)
- **TB** Beneficial effects of oral contraceptives: lower risk of endometrial and ovarian cancer, ovarian cysts and benign breast disease; lifetime breast cancer risk unchanged. p. 773; p. 774

- **TB** Mifepristone (RU-486) blocks progesterone and glucocorticoid receptors; 600 mg with misoprostol ends early pregnancy in over 95%. p. 775

Step by step

Choosing hormonal contraception S5; S9; S45

Screen: VTE/stroke history, smoker > 35, migraine with aura, estrogen-sensitive cancer, uncontrolled HTN

→ Any present → progestin-only (mini-pill, DMPA, implant, LNG IUD) →

None → combined pill, patch or ring (cycle control, acne) →

On rifampin or other inducers → backup method

Figure 1 · Unopposed estrogen: why a progestin is added. Schematic (illustrative) — Figure 1. Progestin counters estrogen's proliferative effect on the endometrium: without it, hyperplasia and cancer risk; estrogen alone only after hysterectomy. S7; S7 (notes)

Figure 2 · The HPG axis: sites of hormonal drug action. Original illustrative diagram (not from Katzung): GnRH → FSH/LH → gonadal sex steroids, which feed back negatively on the hypothalamus and pituitary; GnRH analogs, combined contraceptives, SERMs, aromatase inhibitors and antiandrogens act at different points. S8; S8 (notes)

Combined vs. progestin-only contraception (S9)

	Combined (estrogen + progestin)	Progestin-only
Forms	Oral pills, transdermal patch, vaginal ring	Mini-pill, DMPA injection, subdermal implant, levonorgestrel IUD
How	Suppresses ovulation via negative feedback on FSH/LH	Thickens cervical mucus; counters endometrial proliferation
VTE	Carries estrogen-related VTE risk — screen for contraindications	No estrogen — safe with VTE history, migraine with aura, smokers over 35
Other	Cycle regulation, acne/hirsutism improvement	Irregular bleeding early; DMPA reversible bone loss

How to tell them apart. Sources: S9

Estrogen, progestin and contraceptive profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Combined oral contraceptives (ethinyl estradiol plus a progestin)	**Estrogen plus progestin**	**Tablets (21/28-day packs); patch; vaginal ring**	**Ethinyl estradiol 20-35 mcg plus a progestin PO daily; patch weekly for 3 weeks; ring 3 weeks in, 1 week out**	**Contraception, cycle control, dysmenorrhea, acne**

Drug	Class	Formulation	Adult dose	Indications
Norethindrone (progestin-only pill)	**Progestin**	**Tablet 0.35 mg**	**0.35 mg PO daily at the same time, with no pill-free days**	**Contraception when estrogen is contraindicated, breastfeeding**
Medroxyprogesterone depot (Depo-Provera)	**Injectable progestin**	**IM 150 mg/mL; SC 104 mg/0.65 mL**	**150 mg IM (or 104 mg SC) every 12-13 weeks**	**Contraception, endometriosis pain, endometrial protection**
Levonorgestrel IUD (Mirena)	**Intrauterine progestin**	**Intrauterine system 52 mg (about 20 mcg/day)**	**One system placed in the uterus; effective up to 8 years for contraception**	**Contraception, heavy menstrual bleeding, endometrial protection**
Estradiol (Estrace, Vivelle-Dot)	**Estrogen**	**Oral tablets 0.5/1/2 mg; transdermal patch 0.025-0.1 mg/day; vaginal cream, tablet or ring; injectable valerate**	**Menopausal symptoms: 1-2 mg PO daily or patch 0.05 mg/day twice weekly, lowest effective dose; add a progestin if the uterus is intact**	**Menopausal vasomotor symptoms, genitourinary syndrome of menopause, hypoestrogenism, prevention of osteoporosis (when other agents unsuitable)**

How to tell them apart. Sources: S29; S30; S31; S32; S33

Estrogen, progestin and contraceptive profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Combined oral contraceptives (ethinyl estradiol plus a progestin)	**Nausea, breast tenderness, breakthrough bleeding, headache**	**VTE or stroke history, smoker 35 or older, migraine with aura, estrogen-sensitive cancer, uncontrolled hypertension, liver disease**	**VTE, stroke and MI risk; hyperkalemia with drospirenone; rifampin and some antiepileptics lower efficacy**
Norethindrone (progestin-only pill)	**Irregular bleeding, headache, acne, breast tenderness**	**Breast cancer, undiagnosed vaginal bleeding, liver tumor**	**Efficacy falls if a dose is more than 3 hours late; ectopic pregnancy risk if it fails**
Medroxyprogesterone depot (Depo-Provera)	**Irregular bleeding, weight gain, delayed return of fertility**	**Breast cancer, pregnancy, unexplained vaginal bleeding**	**Boxed warning: loss of bone mineral density; limit prolonged use; VTE**
Levonorgestrel IUD (Mirena)	**Irregular spotting, amenorrhea, ovarian cysts, pelvic pain**	**Pregnancy, pelvic infection, uterine anomaly, breast cancer**	**Uterine perforation, expulsion, infection; check strings**

Drug	Adverse effects	Contraindications	Toxicity / overdose
Estradiol (Estrace, Vivelle-Dot)	**Breast tenderness, nausea, headache, VTE, stroke, endometrial hyperplasia (unopposed), gallbladder disease**	**History of VTE or stroke, estrogen-sensitive cancer, undiagnosed vaginal bleeding, active liver disease, pregnancy**	**Nausea, vomiting, withdrawal bleeding; long-term risk of VTE, stroke, breast cancer; endometrial cancer if unopposed**

How to tell them apart. Sources: S29; S30; S31; S32; S33

WHY IT HAPPENS

- Why a progestin with estrogen when the uterus is intact? Estrogen alone keeps the lining growing — hyperplasia, then cancer; the progestin matures it. S6; S7

HIGH-YIELD

- Estrogen = VTE (smoking, age > 35, VTE history); unopposed estrogen = endometrial cancer → always pair with a progestin if the uterus is intact. S5; S6
- Estrogen contraindicated → progestin-only is the right first choice; DMPA = reversible bone loss. S6; S9

2 PART 2 · DRUG CLASS

SERMs, Aromatase Inhibitors & Androgens

SO2 SO3 IO6.1 IO6.2 IO6.3 IO6.5

CLASS

- **Tissue-selective and blocking strategies — how drugs can turn hormone signaling on in one tissue and off in another.** S10
- A receptor-blocking strategy and a synthesis/activation-blocking strategy exist on both the estrogen and androgen sides. S17; S17 (notes)

MECHANISM OF ACTION

- **Mechanism: bind the estrogen receptor, but act as an agonist in some tissues and an antagonist in others — the defining feature of this class.** Same receptor, different conformation and co-regulators in each tissue. S11; S12 (notes); SG U6-R5
- Tamoxifen: ER antagonist in breast tissue but an ER agonist in bone (preserves density) and, notably, in the endometrium. Tamoxifen antagonizes breast and agonizes bone and endometrium. S11; S12; S10 (notes)
- Raloxifene: ER antagonist in breast and ER agonist in bone, but — unlike tamoxifen — essentially neutral at the endometrium; tamoxifen is an ER agonist at the endometrium (raising endometrial cancer risk), both being ER antagonists in breast. S11; S12; SG U6-R5

PROTOTYPE + OTHER DRUGS NAMED

- Tamoxifen, raloxifene (SERMs); anastrozole, letrozole (aromatase inhibitors); testosterone; spironolactone, flutamide, bicalutamide (antiandrogens); finasteride, dutasteride (5-ARIs). S11; S14; S15

INDICATIONS

- Tamoxifen: ER-positive breast cancer treatment and prevention. Raloxifene: preferred option for osteoporosis prevention in postmenopausal women without a breast cancer indication. S11
- Aromatase inhibitors block peripheral enzymatic conversion of androgens to estrogen rather than acting at the receptor — postmenopausal use only: before menopause the ovaries are the dominant estrogen source (often combined with ovarian suppression). S13 (notes); S14; S14 (notes); SG U6-R6
- ● **Testosterone (androgen):** used for male hypogonadism. S15
- ● **Spironolactone (antiandrogen):** blocks the androgen receptor (in addition to its aldosterone-antagonist diuretic effect from Unit 5) — used off-label for hirsutism and acne in women, especially in PCOS. S15
- ● **Flutamide / bicalutamide (antiandrogens):** pure androgen receptor blockers used primarily in prostate cancer, often combined with a GnRH agonist (leuprolide). S15; S45

- Aromatase chain (Figure 5): androgens (androstenedione, testosterone) → aromatase in fat tissue converts them to estrogens → estrogens drive ER-positive tumor growth → anastrozole and letrozole block aromatase. S14
- **Figure 4. Original illustrative diagram (not from Katzung): SERMs block the estrogen receptor directly in breast tissue (while remaining agonists elsewhere), while aromatase inhibitors block the peripheral enzymatic conversion of androgens to estrogen —** S13
- **two distinct strategies converging on the same clinical goal in hormone-sensitive breast cancer.** Receptor blockade versus synthesis blockade. S13; S13 (notes)
- Androgen pathway (Figure 6): testosterone (circulating androgen) → 5-alpha-reductase converts it to DHT (more potent) → binds the androgen receptor → hair follicle and prostate effects. S16; S15 (notes)
- Finasteride and dutasteride block 5-alpha-reductase, lowering DHT; spironolactone, flutamide and bicalutamide block the androgen receptor. Blocking the enzyme or the receptor both reduce androgen action — different mechanisms grouped under the same broad "antiandrogen" label. S16; S16 (notes); SG U6-R8

- ● **Finasteride / dutasteride (5-alpha-reductase inhibitors):** block conversion of testosterone to the more potent dihydrotestosterone (DHT); used for hirsutism and androgenetic alopecia here, and — in Session 2 — for benign prostatic hyperplasia (BPH). S15; S15 (notes)
- Estrogen-pathway drugs are used primarily in hormone-sensitive breast cancer; androgen-pathway drugs for hirsutism, prostate cancer, alopecia, and BPH. S17
- **TB** Tamoxifen as adjuvant therapy cuts contralateral breast cancer by 35%; benefit does not grow beyond 5 years. p. 774
- **TB** Clomiphene, a partial estrogen agonist, blocks hypothalamic feedback to trigger ovulation (mostly in PCOS); multiple pregnancy about 10%. p. 776; p. 777
- **TB** Other inhibitors: danazol suppresses the midcycle LH/FSH surge (endometriosis; hepatic caution); exemestane is an irreversible aromatase inhibitor. p. 775; p. 776

ADVERSE EFFECTS (BOXED WARNINGS FIRST)

- **Tamoxifen's endometrial agonist activity increases the risk of endometrial hyperplasia and endometrial cancer — patients require counseling on abnormal bleeding and routine gynecologic follow-up.** S11
- Both SERMs carry an increased VTE risk, similar in magnitude to estrogen therapy itself. S11; S12
- Aromatase inhibitor adverse effects: bone loss and joint pain. S14; S14 (notes)
- Testosterone: erythrocytosis, hepatotoxicity (oral formulations), and suppression of endogenous testicular function/fertility. S15
- **TB** Androgens & anabolic steroids: masculinization in women, polycythemia; 17-alkylated oral androgens cause cholestasis, peliosis hepatis and liver tumors. p. 779; p. 780
- **TB** Flutamide causes gynecomastia and occasional reversible hepatotoxicity; bicalutamide is given once daily with a GnRH analog to prevent tumor flare. p. 782

CONTRAINDICATIONS

- Spironolactone: avoid in pregnancy (teratogenic risk). S24
- **TB** Androgens are contraindicated in pregnant women or women who may become pregnant (fetal masculinization). p. 780

DRUG INTERACTIONS

- CYP2D6 inhibitors (paroxetine, fluoxetine) blunt tamoxifen activation. Testosterone and tamoxifen raise INR with warfarin. S45; S45 (notes)
- Bisphosphonates and calcium with vitamin D protect bone with aromatase inhibitors. ACE inhibitors, ARBs and potassium: hyperkalemia with spironolactone. S45; S45 (notes)

PHARMACOKINETICS

- **TB** Androgen suppression: continuous (not pulsatile) GnRH analogs such as leuprolide suppress the gonads; dutasteride has a much longer half-life than finasteride. p. 781

MONITORING

- Testosterone therapy requires monitoring hematocrit (erythrocytosis risk) and PSA with long-term use, with additional risks of hepatotoxicity (oral formulations) and suppression of endogenous fertility. S15; SG U6-R7

CLINICAL PEARL / WHY

- The tamoxifen-vs-raloxifene endometrial distinction is a classic high-yield comparison: tamoxifen carries the endometrial cancer risk because of tissue-selective agonism, not a class effect. S11 (notes); S12 (notes)
- Finasteride/dutasteride are a direct bridge to Session 2 — the same drugs reappear for BPH (prostate growth rather than hirsutism). S15 (notes)
- Case: a 52-year-old postmenopausal woman with ER-positive breast cancer starts tamoxifen; six months later, new-onset irregular vaginal bleeding. **What is the most likely explanation, and what should happen next?** S18
- **Tamoxifen acts as an estrogen receptor agonist in the endometrium even while acting as an antagonist in breast tissue — this tissue-selective agonism increases the risk of endometrial hyperplasia and endometrial cancer.** S18
- **New abnormal bleeding on tamoxifen warrants prompt gynecologic evaluation, not reassurance.** A natural contrast with raloxifene's neutral endometrial profile. S18; S18 (notes)

Step by step

Reducing estrogen signaling in breast cancer S11; S14; S18; S45

ER-positive breast cancer → Premenopausal → tamoxifen (AI alone insufficient)
→ Postmenopausal → aromatase inhibitor or tamoxifen →
Tamoxifen + new bleeding → gynecologic evaluation →
AI → protect bone (calcium, vitamin D, bisphosphonate)

Figure 3 · SERMs: one receptor, different tissue effects. Schematic (illustrative) — Figure 3. Tamoxifen and raloxifene act on the same receptor but differ by tissue; tamoxifen’s endometrial agonism carries cancer risk. Both raise VTE risk. S12; S12 (notes)

Figure 4 · Two ways to reduce estrogen signaling. Original illustrative diagram (not from Katzung): SERMs block the receptor in breast; aromatase inhibitors block peripheral conversion of androgens to estrogen — two strategies for hormone-sensitive breast cancer. S13; S13 (notes)

Figure 5 · Aromatase inhibitors: blocking estrogen synthesis. Schematic (illustrative) — Figure 5. Aromatase converts androgens to estrogens in peripheral fat; aromatase inhibitors block this step. They act downstream of the ovary, so they work mainly after menopause. S14; S14 (notes)

Figure 6 · Androgen pathway: where each drug acts. Schematic (illustrative) — Figure 6. Androgen pathway from testosterone to receptor action, with the sites of finasteride, dutasteride (enzyme) and spironolactone, flutamide, bicalutamide (receptor). S16; S16 (notes)

Estrogen-pathway vs. androgen-pathway drugs (S17)

	Estrogen pathway	Androgen pathway
Receptor strategy	SERMs (tamoxifen, raloxifene): tissue-selective agonist/antagonist	Antiandrogens (spironolactone, flutamide): block the androgen receptor
Synthesis strategy	Aromatase inhibitors (anastrozole, letrozole): postmenopausal only	5-alpha-reductase inhibitors (finasteride, dutasteride): block DHT formation
Main uses	Hormone-sensitive breast cancer	Hirsutism, prostate cancer, alopecia, BPH

How to tell them apart. Sources: S17

Tamoxifen vs. raloxifene (S11–S12)

Tissue	Tamoxifen	Raloxifene
Breast	Antagonist	Antagonist
Bone	Agonist	Agonist
Endometrium	Agonist — hyperplasia/cancer risk	Essentially neutral
VTE	Increased	Increased

How to tell them apart. Sources: S11; S12

SERM and aromatase inhibitor profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Tamoxifen (Nolvadex)	**Selective estrogen-receptor modulator**	**Tablet 10/20 mg**	**20 mg PO daily for 5-10 years in breast cancer**	**Breast cancer treatment and risk reduction, DCIS**
Raloxifene (Evista)	**Selective estrogen-receptor modulator**	**Tablet 60 mg**	**60 mg PO daily**	**Postmenopausal osteoporosis treatment and prevention, breast cancer risk reduction**
Anastrozole (Arimidex)	**Aromatase inhibitor**	**Tablets 1 mg**	**1 mg PO once daily, usually for 5 years or more**	**Adjuvant and metastatic hormone receptor-positive breast cancer in postmenopausal women**
Letrozole (Femara)	**Aromatase inhibitor**	**Tablets 2.5 mg**	**2.5 mg PO once daily, up to 5 years or extended therapy**	**Hormone receptor-positive breast cancer in postmenopausal women; off-label ovulation induction**

How to tell them apart. Sources: S34; S35; S36; S37

SERM and aromatase inhibitor profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Tamoxifen (Nolvadex)	Hot flashes, vaginal discharge, menstrual irregularity	History of DVT or PE, pregnancy, warfarin (for risk-reduction use)	Endometrial cancer, VTE, stroke, cataracts; CYP2D6 inhibitors (paroxetine, fluoxetine) reduce activation
Raloxifene (Evista)	Hot flashes, leg cramps, peripheral edema	VTE history, pregnancy	Boxed warning: VTE and fatal stroke risk; does not stimulate the endometrium
Anastrozole (Arimidex)	Hot flashes, joint pain, bone loss, osteoporosis, mood changes, hypercholesterolemia	Pregnancy; ineffective in premenopausal women unless ovarian function is suppressed	Bone loss and fractures; monitor bone density and lipids; overdose causes mainly GI upset
Letrozole (Femara)	Hot flashes, arthralgia, osteoporosis, fatigue, hypercholesterolemia	Pregnancy; premenopausal status without ovarian suppression	Bone loss and fractures; monitor bone density; nonspecific toxicity in overdose

How to tell them apart. Sources: S34; S35; S36; S37

Androgen and antiandrogen profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Testosterone (cypionate, gel, patch)	Androgen	IM cypionate or enanthate 100-200 mg/mL; gel; patch; pellets	Cypionate 100-200 mg IM every 2 weeks (or 50-100 mg weekly); gel 50-100 mg daily	Male hypogonadism, delayed puberty
Spironolactone (Aldactone)	Aldosterone antagonist with antiandrogen activity	Tablets 25/50/100 mg	Hirsutism or acne (off-label): 50-200 mg PO daily in divided doses; hypertension and heart failure per cardiology dosing	Hirsutism and acne in PCOS (off-label), resistant hypertension, heart failure, primary hyperaldosteronism
Flutamide (Eulexin)	Nonsteroidal antiandrogen	Capsules 125 mg	250 mg PO three times daily, given with a GnRH agonist in prostate cancer	Metastatic prostate cancer in combination with a GnRH agonist
Bicalutamide (Casodex)	Nonsteroidal antiandrogen	Tablets 50 mg (150 mg in some regions)	50 mg PO once daily with a GnRH agonist in prostate cancer	Metastatic prostate cancer in combination with a GnRH agonist

Drug	Class	Formulation	Adult dose	Indications
Finasteride (Proscar, Propecia)	**5-alpha-reductase inhibitor**	**Tablet 5 mg (BPH); 1 mg (hair loss)**	**5 mg PO daily for BPH; 1 mg PO daily for male-pattern hair loss**	**BPH, androgenetic alopecia**
Dutasteride (Avodart)	**5-alpha-reductase inhibitor (types 1 and 2)**	**Capsules 0.5 mg; combination with tamsulosin (Jalyn)**	**0.5 mg PO once daily for BPH; effects take 3-6 months; off-label for hair loss**	**Benign prostatic hyperplasia, off-label androgenetic alopecia and hirsutism**

How to tell them apart. Sources: S38; S39; S40; S41; S42; S43

Androgen and antiandrogen profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Testosterone (cypionate, gel, patch)	**Acne, erythrocytosis, sleep apnea, gynecomastia, testicular atrophy**	**Prostate or breast cancer, polycythemia, pregnancy**	**Erythrocytosis, thrombosis, secondary exposure with gels; monitor hematocrit, PSA, lipids**
Spironolactone (Aldactone)	**Hyperkalemia, menstrual irregularity, breast tenderness, dizziness, gynecomastia in men**	**Hyperkalemia, Addison disease, anuria, pregnancy (feminization of a male fetus), concurrent potassium supplements**	**Life-threatening hyperkalemia, especially with CKD or ACE inhibitors or ARBs; monitor potassium; effective contraception needed**
Flutamide (Eulexin)	**Hot flashes, gynecomastia, diarrhea, hepatotoxicity, methemoglobinemia**	**Severe hepatic impairment, pregnancy**	**Boxed warning for severe hepatotoxicity; monitor liver tests regularly**
Bicalutamide (Casodex)	**Hot flashes, gynecomastia, breast tenderness, fatigue, liver enzyme elevation**	**Pregnancy (not indicated in women), severe hepatic impairment**	**Hepatotoxicity (rare); monitor liver tests; potentiates warfarin**
Finasteride (Proscar, Propecia)	**Decreased libido, erectile dysfunction, reduced ejaculate, gynecomastia**	**Pregnancy and women who may be pregnant (do not handle crushed tablets), hypersensitivity**	**Lowers PSA by about half; reports of depression; teratogenic to a male fetus**
Dutasteride (Avodart)	**Decreased libido, erectile dysfunction, ejaculation disorders, breast tenderness, depression**	**Pregnancy and women of childbearing potential handling capsules, severe hepatic impairment**	**Lowers PSA by about half; pregnant partners must avoid handling capsules; monitor PSA and mood**

How to tell them apart. Sources: S38; S39; S40; S41; S42; S43

WHY IT HAPPENS

- Why are aromatase inhibitors useless before menopause? The ovary still makes most of the estrogen; the drug only blocks the peripheral (fat) source. S13 (notes); S14

HIGH-YIELD

- Tamoxifen: breast antagonist, endometrial agonist → bleeding needs evaluation; raloxifene: endometrium neutral; both VTE. S11; S18
- Aromatase inhibitors: postmenopausal only; bone loss and joint pain. S14

3 PART 3 · DRUG CLASS

PCOS & the Reproductive-Metabolic Overlap

SO4 SO5 IO6.1 IO6.3 IO6.4 IO6.5

CLASS

- **Where Chapter 40 and Chapter 41 meet — insulin resistance as a driver of reproductive endocrine disease.** S19
- **Polycystic ovary syndrome (PCOS) is defined by a combination of hyperandrogenism, ovulatory dysfunction, and polycystic ovarian morphology — but its pharmacologic management is fundamentally a metabolic problem as much as a reproductive one.** S20
- **This is why PCOS management routinely pulls drugs from Chapter 41 (antidiabetic agents) as well as Chapter 40 (hormonal agents) — the two chapters converge on the same patient.** PCOS pharmacology is not purely "gynecologic." S20; S20 (notes)

PROTOTYPE + OTHER DRUGS NAMED

- Metformin; combined hormonal contraception; spironolactone; GLP-1 receptor agonists. S24

MECHANISM OF ACTION

- Insulin resistance and compensatory hyperinsulinemia are central to PCOS pathophysiology in most patients, independent of body weight. S20
- Hyperinsulinemia stimulates ovarian theca-cell androgen production directly and lowers hepatic sex hormone-binding globulin (SHBG) production, raising free (biologically active) testosterone — the mechanistic core of PCOS. S20; S19 (notes); SG U6-R9

INDICATIONS

- Clinical features follow the mechanism: hirsutism, acne, and anovulation/irregular cycles from hyperandrogenism; an elevated long-term risk of type 2 diabetes and metabolic syndrome from insulin resistance. S20
- **Metformin: first-line insulin-sensitizing agent in PCOS management; improves insulin sensitivity, can restore ovulatory cycles, and is commonly used alongside or instead of hormonal therapy depending on the patient's goals (cycle regulation vs. fertility).** S22

- **Figure 7. Original illustrative diagram (not from Katzung): insulin resistance and hyperandrogenism reinforce one another in a self-sustaining cycle — hyperinsulinemia raises free testosterone, and hyperandrogenism worsens insulin resistance,** S21
- **explaining why metabolic therapy improves reproductive symptoms and vice versa.** Walk both arrows of the cycle. S21; S21 (notes)
- Metformin is not a direct hormonal agent — it improves hyperandrogenism indirectly, by reducing hyperinsulinemia's stimulatory effect on ovarian androgen production (Figure 7) — by improving insulin sensitivity, not by acting on any hormone receptor. S22; SG U6-R10
- Chain (Figure 8): insulin resistance → high insulin → ovarian androgens up; liver SHBG down → free testosterone up → metformin lowers insulin → androgens fall; cycles and ovulation can return. S23; S23 (notes)

- GLP-1 receptor agonists are increasingly used off-label in PCOS patients with obesity, given their weight-loss and insulin-sensitizing effects — an emerging extension of the same principle. S22; S45
- Combined hormonal contraception remains first-line specifically for cycle regulation and hirsutism/acne when pregnancy is not desired — metformin and hormonal therapy are frequently used together, not as alternatives. S22; S24 (notes)
- **TB** Polycystic ovary syndrome affects about 7% of women of reproductive age. p. 776

ADVERSE EFFECTS (BOXED WARNINGS FIRST)

- **Adverse effects: GI upset (common, often transient), and rare but serious lactic acidosis risk in renal impairment — recall the renal dose-adjustment principles from Unit 5.** S22; S22 (notes)
- Common early effect: GI upset (nausea, diarrhea); take with meals. S23

CONTRAINDICATIONS

- Hold metformin for an iodinated contrast study or renal impairment. S23; S23 (notes)

DRUG INTERACTIONS

Not covered in materials

PHARMACOKINETICS

Not covered in materials

MONITORING

Not covered in materials

CLINICAL PEARL / WHY

- Case: a 24-year-old with PCOS — irregular cycles, hirsutism, BMI 31 — not trying to conceive, no contraindications to estrogen. **What are the two major pharmacologic strategies available, and how do their goals differ?** S25
- **Combined hormonal contraception directly regulates cycles and improves hirsutism/acne through its hormonal effects, while metformin addresses the underlying insulin resistance driving the condition metabolically.** S25
- **The two are frequently combined rather than chosen as alternatives — contraception for immediate symptom control, metformin for the metabolic driver and long-term diabetes risk reduction.** S25; S25 (notes)
- Drug names reveal the class: **Stem** · **Class** · **Examples** · **Memory aid** — estr-/-estradiol estrogen (estradiol, ethinyl estradiol); -gest-/-gestrel progestin (levonorgestrel, norethindrone, medroxyprogesterone: gest- = pregnancy hormone). S26; S26 (notes)
- -ifene SERM (tamoxifen, raloxifene: modulates the estrogen receptor); -trozole aromatase inhibitor (anastrozole, letrozole: blocks aromatase); -steride 5-alpha-reductase inhibitor (finasteride, dutasteride: blocks steroid enzyme). S26
- -lutamide antiandrogen (flutamide, bicalutamide: blocks the androgen receptor); -sterone androgen or antagonist (testosterone, spironolactone: steroid; drug sets the effect); -formin biguanide (metformin: lowers insulin). S26
- Recap: **1.** Estrogens and progestins are always paired in a patient with a uterus — unopposed estrogen significantly raises endometrial cancer risk (Chapter 40). S27
- **2.** SERMs and aromatase inhibitors are two distinct strategies for reducing estrogen signaling in hormone-sensitive breast cancer — tissue-selective receptor modulation (Figure 3) vs. blocking peripheral synthesis (Figure 5), compared in Figure 4. S27
- **3.** Finasteride and dutasteride appear here for hirsutism and alopecia — the same drugs return in Session 2 for BPH. **4.** PCOS: insulin resistance and hyperandrogenism reinforce one another (Figure 7), so metformin and hormonal therapy are frequently combined. S27; S27 (notes)
- **5.** Session 2 (Tuesday, December 1) shifts to autonomic pharmacology of the bladder and lower urinary tract — mechanistically different, but sharing cross-system drug-class connections with Units 5 and 6. S27; S27 (notes)

Step by step

PCOS loop S23; S25

Insulin resistance → high insulin →

Ovarian androgens up; liver SHBG down →

Free testosterone up: hirsutism, anovulation →

Metformin lowers insulin → androgens fall →

Add combined contraception for cycles and hirsutism

Figure 7 · The PCOS–insulin resistance cycle. Original illustrative diagram (not from Katzung): hyperinsulinemia raises free testosterone (ovarian androgens up, SHBG down) and hyperandrogenism worsens insulin resistance — a self-sustaining cycle that metformin interrupts. S21; S21 (notes)

Figure 8 · Metformin in PCOS: insulin, androgens and ovulation. Schematic (illustrative) — Figure 8. Insulin, androgens and ovulation in PCOS, and where metformin acts in the loop: lower insulin → lower androgens → cycles and ovulation can return. S23; S23 (notes)

Table 1 · Agents used in PCOS management at a glance

Agent	Primary goal	Key teaching point
Combined hormonal contraception	Cycle regulation, hirsutism/acne	Progestin component opposes unopposed-estrogen endometrial risk
Metformin	Insulin sensitization	First-line metabolic therapy; can restore ovulation; GI upset common
Spironolactone	Antiandrogen effect on hirsutism/acne	Androgen receptor blockade; avoid in pregnancy (teratogenic risk)
GLP-1 receptor agonists	Weight loss, insulin sensitization	Emerging off-label use in PCOS with obesity

How to tell them apart. Sources: S24; S24 (notes)

Supporting drugs at a glance (S45)

Drug(s)	Why it matters here
Enzyme inducers	Rifampin, carbamazepine, phenytoin and St. John's wort lower combined contraceptive efficacy
Warfarin	Testosterone and tamoxifen raise INR
GnRH agonists	Leuprolide is used with antiandrogens in prostate cancer
GLP-1 receptor agonists	Off-label for PCOS with obesity
Clomiphene and letrozole	Ovulation induction, beyond this lecture
Bisphosphonates and calcium with vitamin D	Bone protection with aromatase inhibitors
CYP2D6 inhibitors	Paroxetine and fluoxetine blunt tamoxifen activation
ACE inhibitors, ARBs and potassium	Hyperkalemia with spironolactone

How to tell them apart. Sources: S45; S45 (notes)

Metformin profile: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Metformin (Glucophage)	**Biguanide**	**Tablet 500/850/1,000 mg; ER tablet; oral solution**	**Start 500 mg PO daily or BID with meals, titrate to 1,000 mg BID (max 2,550 mg/day)**	**Type 2 diabetes, prediabetes, PCOS (off-label)**

How to tell them apart. Sources: S44

Metformin profile: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Metformin (Glucophage)	**Nausea, diarrhea, metallic taste, vitamin B12 deficiency**	**eGFR below 30, acute metabolic acidosis, hypersensitivity**	**Lactic acidosis (renal impairment, contrast, alcohol, sepsis); hold around iodinated contrast; monitor B12**

How to tell them apart. Sources: S44

WHY IT HAPPENS

- Why does a diabetes drug help hirsutism? Insulin drives the ovary to make androgens and the liver to make less SHBG; lower the insulin and free testosterone falls. S20; S23

HIGH-YIELD

- PCOS = insulin resistance ↔ hyperandrogenism cycle; metformin (metabolic) + combined contraception (cycles, hirsutism) are usually combined. S21; S22; S25

Cram sheet

Every term, mechanism, drug, and number. Hide the answers and quiz yourself; tap a blurred answer to peek.

Prompt	Answer
Estrogen mechanism	ER-α/ER-β → gene transcription S5
Estrogen top AE	VTE (smoking, age > 35) S5
Uterus + estrogen	Always add a progestin S6; S7
Progestin-only methods	Mini-pill, DMPA, implant, LNG IUD — no estrogen VTE risk S6; S9
DMPA	Reversible bone density loss S6; S9
Tamoxifen	Breast −, bone +, endometrium + (cancer); CYP2D6 S11; S45
Raloxifene	Breast −, bone +, endometrium neutral; VTE S11
Aromatase inhibitors	Postmenopausal only; bone loss, joint pain S14
Testosterone	Hematocrit, PSA; oral hepatotoxicity S15
Flutamide/bicalutamide vs finasteride	Receptor block vs DHT synthesis block S16
PCOS core	Hyperinsulinemia ↑ theca androgens, ↓ SHBG S20
Metformin in PCOS	Indirect; GI upset; hold for contrast S22; S23
Spironolactone in PCOS	Androgen receptor block; avoid in pregnancy S15; S24
OC VTE driver	Estrogen dose, not progestin; factor V Leiden ↑ risk p. 772
OC benefits	↓ Endometrial and ovarian cancer p. 773
Clomiphene	Ovulation induction; ~10% multiple pregnancy p. 776; p. 777
Oral 17-alkyl androgens	Cholestasis, peliosis hepatis, liver tumors p. 780

Self-test

First, the lecture's own 0 knowledge checks — exactly as written. Then original PANCE-style questions. Tap an answer to reveal the explanation.

From the lecture

Original questions

Q1 · Clinical vignette · Estrogens, Progestins & Contraception

A 52-year-old with an intact uterus wants estradiol for hot flashes. What must be added?

A	Spironolactone for balance
B	A progestin to protect the endometrium
C	Nothing, if the dose is low
D	Tamoxifen to block the endometrium
E	An aromatase inhibitor

ANSWER: B

Unopposed estrogen with a uterus raises endometrial cancer risk; add a progestin (S6–S7). S6; S7

- **A.** Not the answer (S15).
- **C.** Still unopposed (S6).
- **D.** Tamoxifen stimulates it (S11).
- **E.** Not the answer (S14).

Q2 · Clinical vignette · Estrogens, Progestins & Contraception

A 36-year-old smoker with migraine with aura wants contraception. Which choice fits best?

- **A** Estradiol tablets alone
- **B** A combined vaginal ring
- **C** A combined pill at the lowest dose
- **D** Progestin-only method (LNG IUD)
- **E** A combined transdermal patch

ANSWER: D

Estrogen is contraindicated; progestin-only methods avoid estrogen VTE risk and are the correct first choice (S9). S5; S9

- **A.** Unopposed estrogen (S6).
- **B.** Contains estrogen (S9).
- **C.** Still estrogen (S9).
- **E.** Contains estrogen (S9).

Q3 · Mechanism · Estrogens, Progestins & Contraception

How do progestins protect the endometrium in combined or hormone therapy?

- **A** They thin cervical mucus to aid clearance
- **B** They lower circulating estrogen levels
- **C** They block aromatase in the endometrium
- **D** They block estrogen receptors in the uterus
- **E** They counter estrogen-driven proliferation

ANSWER: E

Progestins bind progesterone receptors, limiting estrogen-driven proliferation and maturing the endometrium (S6–S7). S6; S7; SG U6-R2

- **A.** They thicken mucus (S6).
- **B.** Not the mechanism (S6).
- **C.** Aromatase inhibitors do that elsewhere (S14).
- **D.** They act at PR, not ER (S6).

Q4 · Mechanism · Estrogens, Progestins & Contraception

Which adverse effect is specific to depot medroxyprogesterone (DMPA) with prolonged use?

- A Lactic acidosis in renal impairment
- B Hyperkalemia from potassium retention
- C Reversible bone mineral density loss
- D Erythrocytosis with a high hematocrit
- E Endometrial cancer from unopposed estrogen

ANSWER: C

DMPA specifically causes reversible bone density loss with long-term use (boxed warning) (S6; S9; S31). S6; S9; S31

- **A.** Metformin (S22).
- **B.** Drospirenone or spironolactone (S29; S39).
- **D.** Testosterone (S15).
- **E.** DMPA contains no estrogen (S6).

Q5 · Clinical vignette · Estrogens, Progestins & Contraception

A 30-year-old with factor V Leiden asks for contraception. Which feature of combined pills drives her main risk?

- A Neither; pills lower VTE risk
- B The estrogen component (VTE)
- C Only her age, not the pill
- D The progestin component
- E Only if she also smokes

ANSWER: B

VTE risk relates to the estrogen, not the progestin, and inherited clotting disorders markedly increase it (p. 772). p. 772

- **A.** They raise it (p. 772).
- **C.** Genetic risk matters (p. 772).
- **D.** VTE tracks estrogen (p. 772).
- **E.** Genetic risk alone raises it (p. 772).

Q6 · Mechanism · SERMs, Aromatase Inhibitors & Androgens

Why is raloxifene preferred over tamoxifen for osteoporosis prevention in a woman without breast cancer?

- A It is neutral at the endometrium
- B It does not raise VTE risk
- C It blocks aromatase
- D It is an endometrial agonist
- E It antagonizes bone resorption via DHT

ANSWER: A

Both are bone agonists; raloxifene is essentially neutral at the endometrium (S11). S11; S12

- **B.** Both raise VTE (S11).
- **C.** That is anastrozole (S14).
- **D.** That is tamoxifen (S11).
- **E.** Not the mechanism (S11).

Q7 · Clinical vignette · SERMs, Aromatase Inhibitors & Androgens

A 45-year-old premenopausal woman with ER-positive breast cancer asks about letrozole alone. Key point?

- A It blocks the estrogen receptor
- B Its main risk is endometrial cancer
- C Ideal — works best before menopause
- D It raises estrogen to fight the tumor
- E Insufficient alone before menopause

ANSWER: E

Before menopause the ovaries are the dominant estrogen source; AIs need ovarian suppression (S13–S14). S13; S14

- **A.** It blocks synthesis (S14).
- **B.** Bone loss and joint pain (S14).
- **C.** Postmenopausal only (S14).
- **D.** Lowers it (S14).

Q8 · Mechanism · SERMs, Aromatase Inhibitors & Androgens

How do finasteride and spironolactone differ in reducing androgen action?

- A Both block 5-alpha-reductase
- B Spironolactone lowers testosterone production in the testis
- C Finasteride blocks aromatase
- D Finasteride blocks DHT synthesis; spironolactone the receptor
- E Both block the androgen receptor

ANSWER: D

5-ARIs block testosterone → DHT; spironolactone blocks the androgen receptor (S16). S15; S16

- **A.** Spironolactone blocks the receptor (S16).
- **B.** Receptor blocker (S16).
- **C.** Anastrozole does (S14).
- **E.** Finasteride blocks the enzyme (S16).

Q9 · Clinical vignette · SERMs, Aromatase Inhibitors & Androgens

A man on testosterone therapy has a hematocrit of 55%. What adverse effect is this?

- A Dehydration from spironolactone
- B Hemolysis from G6PD deficiency
- C Erythrocytosis from testosterone
- D Expected; no monitoring needed
- E Aromatase-inhibitor bone loss

ANSWER: C

Testosterone causes erythrocytosis; monitor hematocrit and PSA (S15). S15; S38

- **A.** Not given (S15).
- **B.** Not testosterone (S15).
- **D.** Monitor hematocrit (S15).
- **E.** Not related (S14).

Q10 · Mechanism · PCOS & the Reproductive-Metabolic Overlap

How does metformin improve hirsutism and ovulation in PCOS?

- **A** It blocks aromatase in fat
- **B** Lowers insulin, so ovarian androgens fall
- **C** It blocks 5-alpha-reductase
- **D** It blocks the androgen receptor
- **E** It suppresses FSH and LH directly

ANSWER: B

Metformin reduces hyperinsulinemia’s stimulation of ovarian androgens and raises SHBG — indirect, not hormonal (S22–S23). S22; S23

- **A.** Anastrozole (S14).
- **C.** Finasteride (S16).
- **D.** Spironolactone (S15).
- **E.** Combined contraception (S9).

Q11 · Mechanism · PCOS & the Reproductive-Metabolic Overlap

In PCOS, how does hyperinsulinemia raise free testosterone?

- **A** It raises SHBG, freeing testosterone
- **B** It suppresses LH release entirely
- **C** It blocks aromatase in the ovary
- **D** More ovarian androgen and less SHBG
- **E** It activates 5-alpha-reductase in skin

ANSWER: D

Hyperinsulinemia stimulates theca-cell androgen production and lowers hepatic SHBG (S20). S20; SG U6-R9

- **A.** SHBG falls (S20).
- **B.** Not the mechanism (S20).
- **C.** Not the mechanism (S20).
- **E.** Not the taught mechanism (S20).

Q12 · Clinical vignette · PCOS & the Reproductive-Metabolic Overlap

A woman with PCOS on spironolactone for hirsutism plans pregnancy. What is the key counseling point?

- A Switch to testosterone
- B Continue; it is safe in pregnancy
- C Stop it — teratogenic in pregnancy
- D Double it to improve fertility
- E Add potassium supplements

ANSWER: C

Spironolactone: avoid in pregnancy (teratogenic risk; feminization of a male fetus) (S24; S39). S24; S39

- **A.** Wrong direction (S15).
- **B.** Avoid (S24).
- **D.** Not the advice (S24).
- **E.** Hyperkalemia risk (S45).

Objectives checklist

Tick only what you could teach out loud. Saved on this device.

☐ **IO6.1** Identify major pharmacologic classes used in the management of genitourinary and reproductive conditions and describe their mechanisms of action.

☐ **IO6.2** Describe the indications, contraindications, adverse effects, and clinically significant interactions associated with medications used to treat genitourinary and reproductive conditions.

☐ **IO6.3** Compare pharmacologic agents within and across therapeutic classes to distinguish clinically relevant differences in efficacy, safety, and use.

☐ **IO6.4** Select appropriate pharmacologic therapy for common genitourinary and reproductive conditions based on clinical presentation and patient-specific factors.

☐ **IO6.5** Identify appropriate dosing, administration, and monitoring considerations for medications used in genitourinary and reproductive conditions.

☐ **IO6.6** Explain important medication-use and safety information to patients receiving genitourinary and reproductive pharmacotherapy.

☐ **SO1** Describe the physiologic roles and clinical uses of estrogens and progestins, including combined and progestin-only hormonal contraception.

☐ **SO2** Describe the mechanism, tissue-selective actions, and adverse effects of selective estrogen receptor modulators (SERMs) and aromatase inhibitors.

☐ **SO3** Describe the clinical uses, adverse effects, and monitoring considerations for androgens and antiandrogens.

☐ **SO4** Explain the pathophysiology of polycystic ovary syndrome (PCOS) as a reproductive-metabolic overlap condition, including the central role of insulin resistance.

☐ **SO5** Describe the role of metformin and other antidiabetic agents in the management of PCOS and related reproductive-metabolic conditions.